# Mattia Fait

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## Education

**BSc in Electronic Engineering** — *Trento, Italy*
*University of Trento* — Present

- Thesis: Working on integrated photonics.
- Focus: Photonics, Digital Logic Design, and Signal Processing.

## Selected Public Projects

**nixos-config** — *Systems & Infrastructure*
*NixOS workstation and local NAS configuration* — Present

- Declarative system configuration for a high-efficiency workstation and local NAS, including Hyprland, networking, storage, containers, development tools, and reproducible operations workflows.
- **Stack:** NixOS, Nix flakes, Bash, Hyprland, Docker, Samba, Syncthing.

**RISC-V** — *Computer Architecture*
*Synthesizable RV32IM/Zicsr CPU core in VHDL* — Present

- Designed a modular single-cycle and five-stage pipelined RISC-V core with hazards, forwarding, interrupts, CSRs, memory-mapped peripherals, simulation, and FPGA synthesis flows.
- **Stack:** VHDL, GHDL, Vivado, RISC-V, FPGA, Make.

**ppse-lab-firmware** — *Embedded Systems*
*RP2040 GPS and environmental telemetry logger* — Present

- Implemented real-time GPS acquisition, telemetry, persistent logging, USB mass storage, OLED UI, and sensor feedback for an RP2040-based embedded system.
- **Stack:** C/C++, RP2040, FreeRTOS, TinyUSB, LittleFS, GNSS, SSD1306.

**ppse-hardware** — *Digital Hardware*
*Embedded-system PCB and electronics design* — Present

- Designed the KiCad schematics and PCB for the companion embedded hardware, integrating the microcontroller, power, user interface, and sensor subsystems.
- **Stack:** KiCad, PCB design, electronics, embedded hardware.

**ald-cgra** — *Computer Architecture*
*Runtime-reconfigurable coarse-grained array on FPGA* — Present

- Developed a 4x4 processing-element array with a UART-controlled host tool, emulator, custom configuration language, simulation, synthesis, and verification flows.
- **Stack:** VHDL, C, FPGA, UART, GHDL, Vivado, Make.

**synthesis** — *Formal Methods*
*Lean 4 formal engineering frontend and semantic IR* — Present

- Built an extensible formal framework for typed engineering models, requirements, refinement, domain theories, and machine-checkable proofs.
- **Stack:** Lean 4, Mathlib, formal verification, semantic IR, Nix.

**retro-games** — *Systems Programming*
*Terminal-based classic games collection* — Present

- Implemented a modular ncurses game suite including Snake, Tetris, Pac-Man, Pong, Breakout, Chess, Minesweeper, and Sudoku.
- **Stack:** C, ncurses, GNU Autotools, Unix, modular systems programming.

## Technical Skills

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|---|---|
| **Formal Verification** | Lean 4, Formal Logic, Metaprogramming |
| **Languages** | C, C++, Rust, Python, Bash, LaTeX |
| **Systems** | Arch Linux (custom ISO), NixOS, Docker |
| **DevOps/Tools** | Git, CI/CD, Neovim (custom config), Makefile |
| **Electronics** | Embedded Systems, PCB Design (KiCad), Photonics |

## LANGUAGES

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|---|---|
| **Italian** | Native |
| **English** | Upper-Intermediate (B2) |
| | Intensive B2 Immersion Program – Oxford, UK (Summer 2018) |
| **German** | Elementary (A2); 8 years of school study |